

TOI-2046b

R-2067

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TOI2046_231102 · created 2026-08-09T22:56:53+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460250.6643 +/- 0.0041 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.158 +/- 0.01
Transit depth (Rp/Rs)^2	2.49 +/- 0.33 [%]
Orbital Inclination (inc)	80.27 +/- 0.9
Airmass coefficient 1 (a1)	1.117 +/- 0.017
Airmass coefficient 2 (a2)	-0.078 +/- 0.021
Scatter in the residuals of the lightcurve fit is	1.56 %
Best Comparison Star	#1 - [140, 394]
Optimal Aperture	3.39
Optimal Annulus	5.46
Transit Duration (day)	0.0847 +/- 0.0073

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.158 ± 0.01 vs 0.1213 (deviation 3.49σ)
Duration fit vs published	0.0847 d vs 0.1033 d (deviation 2.43σ)
Mid-time O–C	12.1 min
Depth SNR	15.0
Residual scatter	1.56 %

Light curve

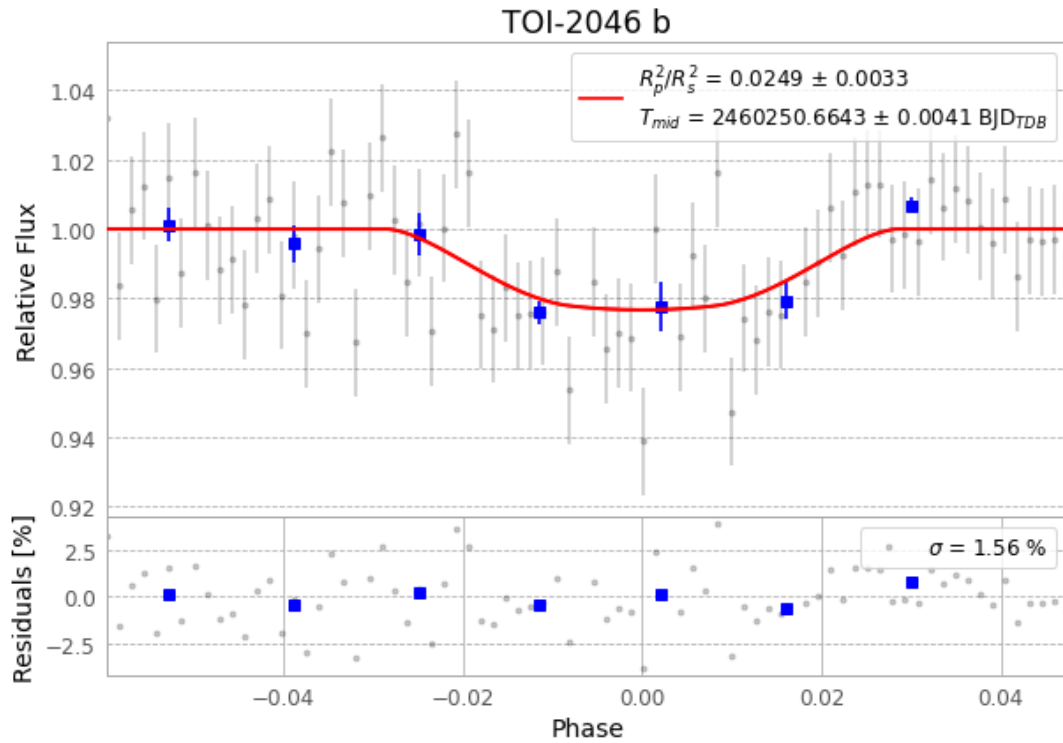


Figure 1 — FinalLightCurve_TOI-2046 b_2023-11-01.png (SHA-256 8da7c75eb1a0b7d6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [305, 248], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9e285e8343f3d3be8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

78 FITS frames (51 MB), TOI2046231102014215.FITS → TOI2046231102053316.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI2046-231102.log (1ae6d56a955b...), FinalLightCurve_TOI-2046 b_2023-11-01.png (8da7c75eb1a0...), FinalLightCurve_TOI-2046 b_2023-11-01.csv (dfa7d17135f3...)

Compute resources

Wall time 142.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-09 22:56Z.